

Discrete Structure (MCA 501)

Chapterwise practice Questions

Unit: 1 Logic and proofs

① Determine whether the following propositions are tautology, Contradiction or contingency.

① $p \vee \neg p$ ② $p \wedge \neg p$ ③ $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$

④ $(q \wedge (p \rightarrow q)) \rightarrow p$ ⑤ $p \vee (q \wedge r)$

② Define inverse, Converse and Contrapositive. Find inverse, Converse and Contrapositive of : "If it rains, then the road is wet".

③ Show the following equivalences

① $p \rightarrow (q \rightarrow p) \leftrightarrow \neg p \rightarrow (p \rightarrow q)$

② $(p \rightarrow q) \wedge (r \rightarrow q) \leftrightarrow (p \vee r) \rightarrow q$

③ $p \rightarrow (q \vee r) \leftrightarrow (p \rightarrow q) \vee (p \rightarrow r)$

④ Determine the validity of following arguments.

If the weather is good, we go for picnic.

If we go for picnic, we enjoy.

Weather is good.

Therefore, we enjoy.

⑤ What is direct proof? Use direct proof to verify that "If n is even, then n^2 is even."

⑥ Using indirect proof and proof by contradiction show that "if n is an integer and $3n+2$ is even, then n is even".

⑦ Using proof by cases show that "if $|n| > 3$, then $n^2 > 9$, where n is a real number.

⑧ Using indirect proof, show that "if $3n+2$ is odd then n is odd".

⑨ What rules of inference is used in each of these arguments?

① prakash is mathematics major. Therefore, prakash is either mathematics major or a computer science major.

② Hari is mathematics major ~~and~~ and a computer science major. Therefore, hari is mathematics major.

③ If it is rainy, then the pool will be closed. It is rainy. Therefore, the pool is closed.

④ If it snows today, the university will closed. The university is not closed today. Therefore, it did not snow today.

⑤ If i go swimming, then i will stay in the sun too long. If i stay in the sun too long, then i will sunburn. Therefore, if i go swimming, then i will sunburn.

⑩ Use rules of inference to show that the hypothesis "sita works hard". "If sita works hard, then she is a dull girl!" and "If sita is a dull girl, then she will not get the job!" imply the conclusion "sita will not get the job".

Unit: 2 Induction and Recursion

① prove by mathematical induction that,
 $1 + 3 + 5 + \dots + (2n-1) = n^2$.

② Using mathematical induction show that
 $1 + a + a^2 + \dots + a^n = \frac{a^{n+1} - 1}{a - 1}$ for $n \geq 1$ and $a \neq 1$.

③ Use mathematical induction to show that "3 divides $n^3 + 2n$ " whenever n is a positive integer.

④ Use mathematical induction to show the inequality $n < 2^n$ for all positive integers $n \geq 1$.

⑤ Use mathematical induction to show that $2^n < n!$ for every positive integer n with $n \geq 4$.

⑥ Define Strong induction. Use strong induction to show that $n^3 - n$ is divisible by 3 whenever n is a positive integer.

⑦ Define recursive definition of a function. Suppose f is defined by

$$f(0) = 3$$

$$f(n+1) = 2f(n) + 3$$

Find $f(1)$, $f(2)$, $f(3)$ and $f(4)$.

⑧ Give a recursive definition for finding factorial and n th term of fibonacci sequence.

Unit: 3 Number Theory

① Define divisibility. Use divisibility to show that if a, b and c are integers. Then

(a) if $a|b$ and $a|c$, then $a|(b+c)$.

(b) if $a|b$, then $a|bc$ for all integers c .

(c) if $a|b$ and $b|c$, then $a|c$.

② Find the first five random numbers using linear congruential method with $x_0 = 29$, $a = 9$, $c = 49$ and $m = 100$.

③ Define GCD and LCM. Use the prime factorization method to find the GCD and LCM of 1000 and 625.

④ Write euclidean algorithm. Find the $\gcd(421, 111)$ using euclidean algorithm.

⑤ Write an algorithm for extended euclidean algorithm. Find the multiplicative inverse of 5 in $\mathbb{Z}_{12}$ using extended euclidean algorithm.

⑥ Express GCD of 124 and 232 in the form of linear combination.

⑦ Find all solutions to the system of congruences using Chinese remainder theorem.

$$x \equiv 1 \pmod{2}$$

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 4 \pmod{11}$$

Unit: 4 Recurrence Relations

- ① Solve the recurrence relation $a_n = a_{n-1} + 6a_{n-2}$ for $n \geq 2$, $a_0 = 3, a_1 = 6$.
- ② What is the solution of the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with initial conditions $a_0 = 1$ and $a_1 = 6$?
- ③ Solve the recurrence relation $a_n = 2a_{n-1} + a_{n-2} - 2a_{n-3}$ for $n \geq 3$, $a_0 = 3, a_1 = 6$ and $a_2 = 9$.
- ④ Solve the recurrence relation $a_n = 5a_{n-1} - 7a_{n-2} + 3a_{n-3}$ for $n \geq 3$, $a_0 = 1, a_1 = 9$ and $a_2 = 15$.
- ⑤ Find all the solutions of the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2} + 7n$.
- ⑥ Find all the solutions of the recurrence relation $a_n = 2a_{n-1} + n \cdot 2^n$.
- ⑦ Find all the solutions of the recurrence relation $a_n = 3a_{n-1} + 2n$. What is the solution with $a_1 = 3$?
- ⑧ Find all the solutions of the recurrence relation $a_n = 4a_{n-1} + n^2$. Also find the solution of the recurrence relation with $a_1 = 1$?
- ⑨ What form does a particular solution of the linear non-homogeneous recurrence relation $a_n = 6a_{n-1} - 9a_{n-2} + F(n)$ when
 - a) $F(n) = 3^n$ b) $F(n) = n \cdot 3^n$ c) $F(n) = n^2 \cdot 2^n$ and
 - d) $F(n) = (n^2 + 1) 3^n$.

Unit: 5 Relation and Graph Theory.

- ① Let $A = \{1, 2, 3\}$ and $B = \{2, 3\}$ are two sets. Find the relation R defined by $R = \{(a, b) \mid a \geq b\}$? Also find Domain and Range of relation R ?
- ② What is relation? Write the different properties of a Relation with example.
- ③ Represent the relation $R = \{(1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (3, 1), (3, 3), (4, 1), (4, 3)\}$ using
 - a) Matrix
 - b) Digraph
- ④ Let $A = \{1, 2, 3\}$ and consider the relation $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$ defined on the set A . Determine whether R is an equivalence relation on A .
- ⑤ Let $A = \{1, 2, 3, 4, 5\}$. Let the relation $R = \{(a, b) \mid a \leq b\}$. Determine if the relation R is partial order relation or not.
- ⑥ Let $A = \{1, 2, 4, 8\}$. Determine if $(A, \mid)$ is poset? Where $\mid$ is divide relation.
- ⑦ What is hasse diagram? Write the different steps to create a hasse diagram. Construct a hasse diagram for the poset $(A, \leq)$, Where Set $A = \{1, 2, 3, 4\}$.
- ⑧ Define the terms order and size of graph, degree of vertex, isolated and pendant vertex with an example.

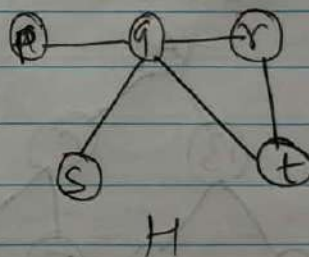
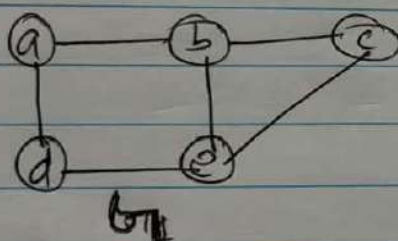
⑨ Define handshaking theorem. How many vertices do the following graphs have

(a) 21 edges, 3 vertices of degree 4 and all other vertices of degree 3.

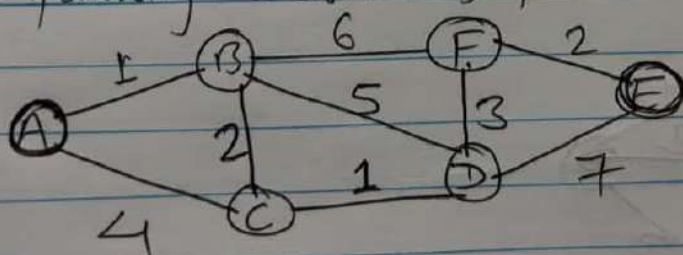
(b) 25 edges, and 2 vertices of degree 5, 3 vertices of degree 4 and other vertices of degree 2.

⑩ Explain Euler and Hamiltonian graph with an example.

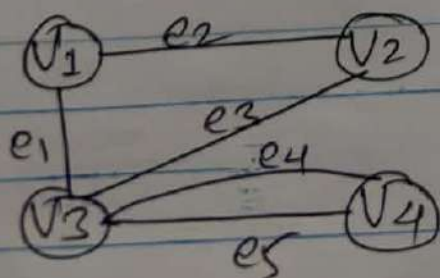
⑪ When two graphs are said to be isomorphic? Show that the following graphs G and H are isomorphic.



⑫ Use Dijkstra's algorithm to find the shortest path from A to E in the following weighted graph.



⑬ Represent the given graph



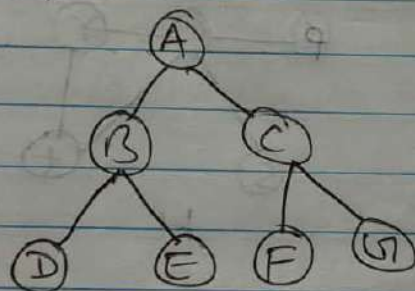
using

- (a) Adjacency matrix.
- (b) Incidence matrix
- (c) Adjacency list

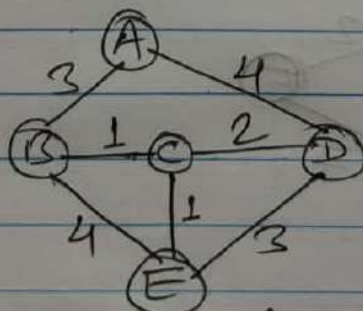
⑭ Suppose that there are six final courses to be scheduled and courses are numbered from C_1 to C_6 . Suppose that the course number $C_1 \& C_2$, $C_1 \& C_3$, $C_2 \& C_3$, $C_3 \& C_4$, $C_4 \& C_5$, $C_4 \& C_6$, $C_5 \& C_6$, $C_2 \& C_6$ has a common students for exam. Then find the final schedule for the exam so that no ~~two~~ students has two exam at the same time?

⑮ Define the terms: binary search tree, game tree, decision tree, Huffman tree and Expression tree with an example.

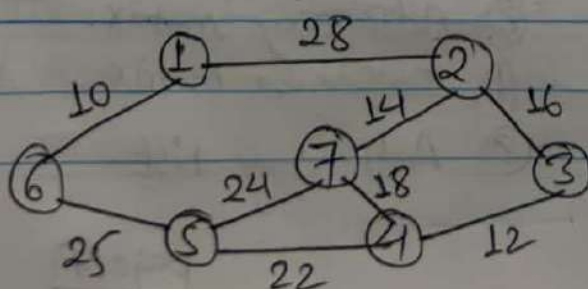
⑯ perform pre-order, in-order and post-order tree traversal for the following binary tree



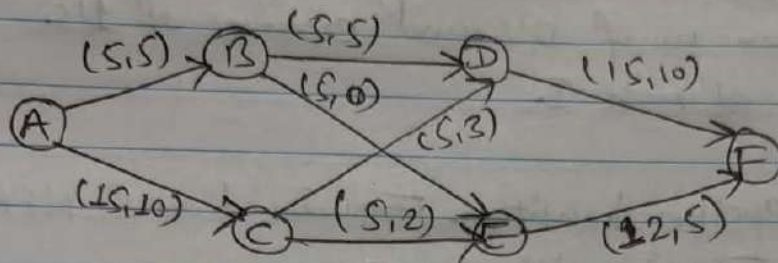
⑰ Use Kruskal's algorithm to find MST.



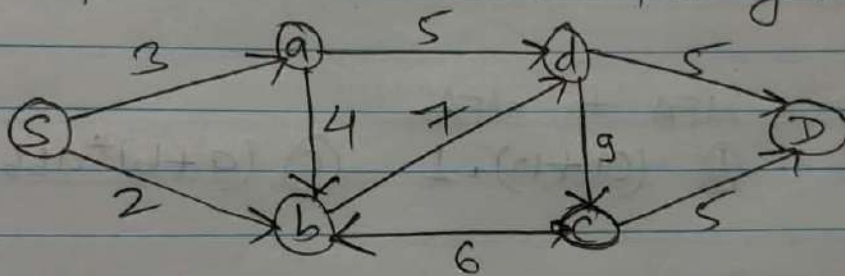
⑱ Use Prim's algorithm to find MST.



- 19) Define maxflow. Find the maximal flow from A to F in the following network flow.



- 20) Define S-D cut and capacity of S-D cut. Calculate the capacity of all possible S-D cuts in the following transport network.



Unit 6 Finite Automata.

- ① Differentiate DFA and NFA with suitable examples.
- ② Draw DFA and NFA for the following languages over $\{0,1\}$
 - a) All strings with even no. of 0's and even no. of 1's.
 - b) All strings of length at most 3.
- ③ Define NFA with epsilon transition. Explain how E-NFA is converted into DFA? Explain with example.
- ④ Define regular expression & Write the different properties of regular expression.
- ⑤ Convert following E-NFA to NFA.
 - a) $(0+1)^*01$
 - b) $(01+10).1$
 - c) $(a+b)^*a5b.$
 - d) $(a+\epsilon).bab$